

ADOMAS VALANTINAS, PHD

Senior Fellow, Leverhulme Centre for Life in the Universe, University of Cambridge, UK

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RESEARCH INTERESTS

I study the surface evolution of rocky planets, asking how water, redox chemistry and climate shaped their early crusts and whether those environments could have supported life, with Mars as my primary focus. I combine orbital remote sensing and machine learning on mission datasets with laboratory experiments on the formation and spectral signatures of iron minerals, and ground both in rover observations and terrestrial analogs.

RESEARCH APPOINTMENTS

LCLU Senior Fellow 09/26–present

Leverhulme Centre for Life in the Universe (LCLU), University of Cambridge, UK

Host: Prof. Nicholas Tosca

Focus: experimental aqueous geochemistry of iron minerals and potential biosignatures on Mars

SNSF Postdoc Mobility Fellow 09/23–08/26

Brown University, USA (two years) and ETH Zurich, Switzerland (one year)

Advisors: Prof. Jack Mustard (Brown) and Prof. Cara Magnabosco (ETH)

Focus: mapping habitable environments on ancient Mars through aqueous mineralogy and geochemistry

Postdoctoral Researcher 12/22–03/23

University of Bern, Switzerland

Advisor: Dr. Antoine Pommerol

Focus: iron oxides in Martian dust as records of ancient aqueous chemistry (laboratory spectroscopy)

Graduate Research Assistant (Ph.D.) 10/18–11/22

University of Bern, Switzerland

Advisor: Prof. Nicolas Thomas

Focus: present-day climate of Mars and related surface processes: dust, frost and slope activity

Master's Degree Researcher 10/15–10/17

University of Copenhagen, Denmark

Advisor: Prof. Kjartan Kinch

Focus: development and characterisation of the Mastcam-Z flight calibration targets now operating on NASA's Perseverance rover

Student Research Assistant (software) 10/15–10/17

Microsoft Station Q and Center for Quantum Devices, Niels Bohr Institute, University of Copenhagen, Denmark

Focus: software development for laboratory instrument control and computer hardware maintenance

EDUCATION

Ph.D. Astronomy and Physics 10/18–11/22

University of Bern, Switzerland

Thesis: *Bright and Dusty Regions of Mars: New Insights from Experiments and Orbital Color Imaging*

Advisor: Prof. Nicolas Thomas

M.Sc. Geophysics 10/15–10/17

University of Copenhagen, Denmark

Advisor: Prof. Kjartan Kinch

B.Sc. Astronomy and Physics <i>Vilnius University, Lithuania</i> Advisor: Dr. Audrius Bridzius	09/11–07/15
Astronomy and Physics (year abroad) <i>University of Glasgow, UK</i>	09/11–06/12

GRANTS AND AWARDS

AI for Science Program award, Anthropic (PI; \$20,000 in Claude API credits)	10/26
LCLU Senior Fellowship, Leverhulme Centre for Life in the Universe, University of Cambridge (PI; \$374,000)	09/26–08/30
NASA Solar System Science program, “Simulating cloud formation on Mars in the laboratory: probing the onset of nucleation in a realistic Martian atmosphere environment”, (Collaborator to PI Dr. Marek Slipski, NASA Jet Propulsion Laboratory; ~\$150,000)	04/26–03/27
Swiss National Science Foundation (SNSF) Postdoc.Mobility return grant, “Potential for ancient habitability and deep crustal materials preserved in the rim of Jezero crater, Mars”, ETH Zurich (PI; \$155,000)	09/25–08/26
Swiss National Science Foundation (SNSF) Postdoc.Mobility Fellowship, “Windows into Mars’ potentially habitable subsurface in the context of NASA’s Perseverance extended mission”, Brown University (PI; \$146,000)	09/23–08/25
American Philosophical Society Baruch Blumberg Grant for Astrobiology, “Decoding Mars Through Iceland: Ferrihydrite as a Key to Past Habitability and Potential Biosignatures”, for field work and conference attendance in Iceland (PI; \$8,500)	07/25
CIFAR seed funding, “Interfaces of Possibility: Modeling Boundary Ecosystems Across Planetary Systems” (Co-PI; \$7,200)	05/25
Lunar and Planetary Institute (LPI) Career Development Award (meeting canceled due to COVID)	03/20
American Geophysical Union (AGU) student travel award (\$2,000)	12/17
Oticon Foundation (Oticon Fonden) grant, Denmark (~\$3,300)	06/16

MISSION EXPERIENCE

NASA Perseverance Rover , PIXL Science Team collaborator (PI: A. Allwood). Contributing to science discussions and operations planning.	09/26–present
ESA Trace Gas Orbiter (TGO) / CaSSIS , Science Team collaborator and operations planner (PI: A. Pommerol). Planned 10 months of CaSSIS observations (~10% of total mission time). Science Theme lead for Hydrothermal Processes and for Geologic Contacts & Stratigraphy. Contributed 31,000 targets to the CaSSIS targeting database (~60% of all team targets).	10/18–present
NASA Mars Reconnaissance Orbiter (MRO) / HiRISE , Science Team collaborator (PI: S. Byrne). Regular contributor to the HiRISE targeting database.	06/20–present
NASA Double Asteroid Redirection Test (DART) , guest participant in the DART Borders program (PI: A. Cheng). Observed team meetings and took part in discussions.	09/22–01/23
NASA Perseverance Rover , Mastcam-Z Science Team collaborator (PI: J. Bell). Developed and characterised the flight calibration targets now operating on Mars.	10/15–10/17

RESEARCH METHODS AND TECHNICAL EXPERTISE

Laboratory spectroscopy: VNIR–SWIR reflectance spectroscopy and spectro-goniometry (analogue soils, dust, frosts); FTIR; Hapke radiative-transfer and intimate-mixture modelling.

Wet-lab: Fe-(oxyhydr)oxide synthesis; planetary regolith analogue synthesis; XRD; experimental aqueous geochemistry of Fe minerals (from 2026).

Orbital remote sensing: Visible imaging (0.4–1.1 μm) — HiRISE/MRO, CTX/MRO, CaSSIS/TGO, HRSC/MEx, MOC/MGS, LROC/LRO. Near-infrared spectroscopy (0.4–5 μm) — CRISM/MRO, OMEGA/MEx, M³/Chandrayaan-1, IIRS/Chandrayaan-2, SP/Kaguya. Thermal infrared (6–50 μm) — THEMIS/Odyssey, TES/MGS, Diviner/LRO. Topography and mapping — MOLA/MGS, HiRISE/CTX stereo DEMs (Ames Stereo Pipeline), crater size–frequency dating, geologic mapping.

Machine learning and data science: Unsupervised clustering and dimensionality reduction (k-means, HDBSCAN, UMAP) and supervised classification of CaSSIS and M³ spectra; deep-learning (CNN) object detection in global orbital image catalogues; zero-/few-shot image classification and feature mapping with multimodal LLMs (Claude API).

Flight hardware and operations: Mastcam-Z/Perseverance calibration targets (development, characterisation); CaSSIS/TGO and HiRISE/MRO targeting and observation planning; PIXL and SuperCam/Perseverance operations (incoming).

Field work: Volcanic, hydrothermal and Fe-rich spring environments sampled for spectral and mineralogical analysis — Iceland, Azores, Canary Islands, Block Island.

FIRST-AUTHOR PUBLICATIONS

1. **Valantinas, A.**, Mustard, J., Tullo, A., Bramble, M.S. and Head, J.W. An exposed magmatic feeder system for the regional olivine unit in Nili Fossae, Mars. Accepted at *Earth and Planetary Science Letters*; preprint on *ESS Open Archive*. doi:10.22541/essoar.176235473.32635438/v4
2. **Valantinas, A.**, Rajsic, A., Mustard, J., Tullo, A. et al. Partially altered olivine in the western rim of Jezero crater, Mars: preserved pre-impact Noachian basement. In review at *GRL*; preprint on *ESS Open Archive*. doi:10.22541/essoar.175579625.58968122/v3
3. **Valantinas, A.**, Mustard, J., Chevrier, V. et al. Detection of ferrihydrite in Martian red dust records ancient cold and wet conditions on Mars. *Nature Communications*, 16, 1712 (2025). doi:10.1038/s41467-025-56970-z
4. **Valantinas, A.**, Thomas, N., Pommerol, A. et al. Evidence for transient morning water frost deposits on the Tharsis volcanoes of Mars. *Nature Geoscience*, 17, 608–616 (2024), cover article. doi:10.1038/s41561-024-01457-7
5. **Valantinas, A.**, Becerra, P., Pommerol, A. et al. CaSSIS color and multi-angular observations of Martian slope streaks. *Planetary and Space Science*, 209, 105373 (2021). doi:10.1016/j.pss.2021.105373
6. **Valantinas, A.** and Schultz, P.H. The origin of neotectonics on the lunar nearside. *Geology*, 48, 649–653 (2020). doi:10.1130/G47202.1
7. **Valantinas, A.**, Kinch, K.M. and Bridzius, A. Low crater frequencies and low model ages in lunar maria: Recent endogenic activity or degradation effects? *Meteoritics & Planetary Science*, 53, 826–838 (2018). doi:10.1111/maps.13033

CO-AUTHOR PUBLICATIONS

8. Bramble, M., (...) **Valantinas, A.** et al. An intrusive origin for the Nili fossae regional olivine-bearing unit. Accepted at *Communications Earth & Environment*.
9. Munaretto, G., (...) **Valantinas, A.** et al. Evidence of mafic silicates in the Blue unit of Phobos: Vis-NIR analogues within the C and Xc-type asteroid taxonomy. In review at *Astronomy & Astrophysics*.

10. Beccarelli, J., (...) **Valantinas, A.** et al. Spectral variability of Phobos and Deimos from TGO/CaSSIS multiband observations. *Astronomy & Astrophysics*, 707, L6 (2026). [doi:10.1051/0004-6361/202558051](https://doi.org/10.1051/0004-6361/202558051)
11. Munaretto, G., (...) **Valantinas, A.** Phase reddening of Phobos and Deimos from TGO/CaSSIS observations. *Astronomy & Astrophysics*, 700, L1 (2025). [doi:10.1051/0004-6361/202555720](https://doi.org/10.1051/0004-6361/202555720)
12. Bickel, V. and **Valantinas, A.** Streaks on Martian slopes are dry. *Nature Communications*, 16, 4315 (2025). [doi:10.1038/s41467-025-59395-w](https://doi.org/10.1038/s41467-025-59395-w)
13. Thomas, N., (...) **Valantinas, A.** et al. Seasonal and short timescale changes on the Martian surface: Multi-spacecraft perspectives. *Space Science Reviews*, 221, 3 (2025). [doi:10.1007/s11214-024-01128-4](https://doi.org/10.1007/s11214-024-01128-4)
14. Mills, M.M., (...) **Valantinas, A.** A global dataset of pitted cones on Mars. *Icarus*, 418, 116145 (2024). [doi:10.1016/j.icarus.2024.116145](https://doi.org/10.1016/j.icarus.2024.116145)
15. Pommerol, A., (...) **Valantinas, A.** et al. In-flight radiometric calibration of the ExoMars TGO Colour and Stereo Surface Imaging System. *Planetary and Space Science*, 223, 105580 (2022). [doi:10.1016/j.pss.2022.105580](https://doi.org/10.1016/j.pss.2022.105580)
16. Munaretto, G., (...) **Valantinas, A.** et al. Multiband photometry of Martian Recurring Slope Lineae (RSL) and dust-removed features at Horowitz crater, Mars from TGO/CaSSIS color observations. *Planetary and Space Science*, 214, 105443 (2022). [doi:10.1016/j.pss.2022.105443](https://doi.org/10.1016/j.pss.2022.105443)
17. Dundas, C.M., (...) **Valantinas, A.** et al. Active Mars: A dynamic world. *JGR: Planets*, 126, e2021JE006876 (2021). [doi:10.1029/2021JE006876](https://doi.org/10.1029/2021JE006876)

MANUSCRIPTS IN PREPARATION

18. **Valantinas, A.**, Girjoaba, A. and Magnabosco, C. Acid Mars, not so acidic? Widespread ferrihydrite–sulfate deposits in CRISM VNIR data. Also, an oral presentation at AGU Fall Meeting 2026. Supported by the Anthropic AI for Science Program.
19. **Valantinas, A.**, Vincendon, M., Pommerol, A., Ottersberg, R., Hauber, E., Tirsch, D. and Robbins, S. Water ice “snowfall” on Martian volcanoes: multi-instrument observations, laboratory experiments and climate modeling.
20. **Valantinas, A.**, Mustard, J., Tosca, N. and Pommerol, A. Mars Global Dust simulant MGD-1: a ferrihydrite-based open standard for fine-grained Martian dust analogues.
21. **Valantinas, A.** and Mustard, J. Evidence for trace Mg-spinel in the Rook-ring anorthosite of Orientale basin from Moon Mineralogy Mapper spectra and Hapke mixture models.
22. **Valantinas, A.** and Tosca, N. Abiotic formation of leopard-spot redox features: implications for potential biosignatures on Mars.
23. Baisch, E.J., D’Andria, M., **Valantinas, A.**, Cooper, T.A. and Warren, K.J. Low-temperature water and preparation-dependent oxygen from heating a ferrihydrite analog of Martian dust. Student-led manuscript; co-supervised.
24. Miller, S.G., (...) **Valantinas, A.** et al. TIDES: A Triton Multiple-Flyby Mission Concept for the New Frontiers Program.

STUDENT SUPERVISION

Elia J. Baisch, M.Sc. student (ETH Zurich, Solar Energy Engineering Laboratory). M.Sc. thesis project on water and oxygen release from heating ferrihydrite-bearing Martian dust analogues; first-author manuscript in preparation. Co-supervisor (main supervisor: Prof. Kent J. Warren).	02/26–present
Luis Grimaldo Estrada, M.Sc. student (ETH Zurich). Machine Learning semester project and M.Sc. thesis project (incoming). Main supervisor.	09/26–present

Ari Antje Hölscher, M.Sc. student (ETH Zurich). Machine Learning for Earth and Planetary Sciences III summer project. Main supervisor; co-supervised with Prof. Cara Magnabosco.	Summer 2026
Chen Orenshtein, M.Sc. student (ETH Zurich). Machine Learning for Earth and Planetary Sciences III summer project. Main supervisor; co-supervised with Prof. Cara Magnabosco.	Summer 2026
Andrei Girjoaba, M.Sc.-level research assistant (ETH Zurich). Machine learning for Mars remote sensing data. Main supervisor; co-supervised with Prof. Cara Magnabosco.	03/26–09/26
Guilliat Corboz, B.Sc. thesis student (ETH Zurich). “Searching for Hydrothermal Systems on Mars: Using Polygonal Ridges to Map Subsurface Habitability”. Main supervisor; co-supervised with Prof. Cara Magnabosco.	02/26–06/26
Rushil Kukreja, high school research mentee (Brown). Spectral analysis of Martian orbital data.	09/24–05/25

TEACHING

Teaching on the MPhil in Planetary Science and Life in the Universe, University of Cambridge (from Michaelmas Term 2026).	10/26–present
Teaching assistant, ETH Zurich. Machine Learning for Earth and Planetary Sciences II & III courses.	02/26–09/26
Laboratory teaching assistant, University of Bern. Instructed students in optics and thermodynamics laboratory courses.	10/18–11/22
Private tutor in physics and mathematics for high school students, Lithuania.	11/17–09/18

PROFESSIONAL SERVICE AND LEADERSHIP

Co-convener and session chair, “The Emergence of Life as a Planetary Process: Prebiotic Chemistry and Habitability on the Early Earth and Beyond”, AGU Fall Meeting 2026.	12/26
Primary convener and session chair, “The Emergence of Life as a Planetary Process”, AGU Fall Meeting 2025.	12/25
Led the Astrobiology and Origin of Life (OLA) reading group at Brown University: biweekly seminars bringing together undergraduates, graduate students, postdocs and faculty from Physics and from Earth, Environmental & Planetary Sciences to discuss seminal books and papers in astrobiology.	01/24–05/25
Panel member, NASA Mars Data Analysis Program (MDAP) review panel.	01/24
Dwornik Award judge for student presentations, Lunar and Planetary Science Conference.	03/23 & 03/25
Peer reviewer for <i>Earth-Science Reviews</i> , <i>Geology</i> , <i>Geophysical Research Letters</i> , <i>Journal of Geophysical Research</i> , <i>Planetary Science Journal</i> and <i>Icarus</i> .	09/21–present
Member, Geological Society of America.	09/21–present
Member, American Geophysical Union.	12/17–present

OUTREACH AND PUBLIC ENGAGEMENT

Dry origin of Martian slope streaks (Bickel & Valantinas, <i>Nature Communications</i> , 2025). Covered by 130 news outlets. Altmetric Attention Score 1,031: top 1% of all research outputs tracked by Altmetric and 7th of 2,532 Nature Communications articles of similar age.	05/25
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Ferrihydrite in Martian dust (Valantinas et al., <i>Nature Communications</i> , 2025). Covered by 187 news outlets; expert interviews with the Associated Press, CNN, The Guardian, the Royal Society of Chemistry and Swiss and Lithuanian national television; guest on the Planetary Society's <i>Planetary Radio</i> podcast. Altmetric Attention Score 1,497: top 1% of all research outputs tracked by Altmetric and 5th of 2,489 Nature Communications articles of similar age; 48,000 article accesses.	02/25
Morning frost on the Tharsis volcanoes (Valantinas et al., <i>Nature Geoscience</i> , 2024; cover article, July 2024). Covered by 311 news outlets; live interviews with the BBC (TV) and ABC Australia (radio), and interviews with CNN, USA Today and New Scientist. Altmetric Attention Score 2,403: 10th of 4,065 Nature Geoscience articles ever tracked and top 1% of all research outputs.	06/24
Radio DJ and science communicator at NTS.live and RadioVilnius.live, curating shows that combine experimental electronic music with planetary science for non-academic audiences.	06/23–present
Artistic collaboration with Matteo Taramelli for the exhibition “Potentially Uninhabitable Worlds”, Dreiviertel Gallery, Bern, Switzerland.	06/21
Lunar neotectonics (Valantinas & Schultz, <i>Geology</i> , 2020). Interview with <i>Ciel et Espace</i> (French popular-science magazine).	06/20

TALKS AND SEMINARS

“From Mariner to Perseverance: 60 Years of Clues in the Search for Martian Life”, Centre for Origin and Prevalence of Life (COPL) seminar, ETH Zurich, Switzerland.	03/26
“Why is Mars red? Recent advances in solving the century-old problem in planetary science”, MIT/Harvard Planetary meeting, MIT, USA.	09/24
“Are Martian volcanoes dormant?”, Planetary Lunch Bunch, Brown University, USA.	05/24
“Neotectonics on the Lunar Nearside”, Observatoire de la Côte d’Azur, Nice, France.	10/17

SELECTED CONFERENCE ABSTRACTS (FIRST AUTHOR)

Selected from 40+ conference abstracts	
Valantinas, A. , Girjoaba, A. and Magnabosco, C. Acid Mars, not so acidic? Widespread ferrihydrite–sulfate deposits in CRISM VNIR data. <i>AGU Fall Meeting 2026</i> , San Francisco (oral).	12/26
Valantinas, A. Unraveling the redox history of Mars. <i>Origins Federation Annual Meeting 2026</i> in Switzerland (poster).	09/26
Valantinas, A. , Mustard, J.F., Tullo, A., Bramble, M.S. and Head, J.W. Evidence for intrusive magmatism as the source of the olivine deposit in the Nili Fossae region, Mars. <i>57th Lunar and Planetary Science Conference</i> , abstract 1470 (oral).	03/26
Valantinas, A. , Vincendon, M., Pommerol, A., Tirsch, D., Hauber, E., Robbins, S.J. and Thomas, N. “Snowfall” on Martian volcanoes. <i>57th Lunar and Planetary Science Conference</i> , abstract 1477 (poster).	03/26
Valantinas, A. , Mustard, J.F., Chevrier, V. et al. Unlocking early Mars redox conditions: evidence from planetary-scale iron oxide deposits. <i>Mars Through Time International Conference</i> , Paris, France (oral).	10/25
Valantinas, A. , Rajsic, A. and Mustard, J.F. Potential for ancient habitability and deep crustal materials preserved in the rim of Jezero crater. <i>56th Lunar and Planetary Science Conference</i> , abstract 2068 (oral).	03/25

Valantinas, A. and Mustard, J.F. Ferrihydrite in dust and bedrock on Mars: astrobiological significance. <i>Astrobiology Science Conference (AbSciCon)</i> , Providence, Rhode Island (oral).	05/24
Valantinas, A. , Mustard, J.F. and Bickel, V.T. ‘Windows’ into Mars’ potentially habitable subsurface. <i>AGU Fall Meeting 2023</i> , San Francisco (oral).	12/23

WORKSHOPS AND SUMMER SCHOOLS

NASA Planetary Science Summer School (PSSS) for mission design, JPL, Pasadena, USA. Helped design a concept mission to Neptune’s moon Triton.	08/25
CIFAR and SNSF “Forum on Radical Interdisciplinarity”, Geneva, Switzerland.	03/25
CIFAR “Earth 4D: Subsurface Science & Exploration” program meeting, Quebec City, Canada.	06/24
European Astrobiology Institute, “Life in the Sub-Surface: Habitats, Species, Metabolism and Survival Strategies”, Azores, Portugal.	10/23
European Space Agency workshop “Mars V”, Les Houches, France.	04/22
GMAP (Geologic Mapping of Planetary Bodies) Planetary Geologic Mapping Winter School (virtual).	01/22
GeoPlaNet analogue field winter school, “Fluid-Rock Interactions in the Solar System”, Nantes, France.	11/18
Nordic Astrobiology Network field summer school, “Impacts and their Role in the Evolution of Life”, Saaremaa, Estonia.	07/17
Nordic Astrobiology Network field summer school, “Biosignatures and the Search for Life on Mars”, Iceland.	06/16

FIELD EXPERIENCE

Iceland: field project funded by the Baruch Blumberg Grant for Astrobiology.	07/25
Block Island, USA: characterization of iron-rich springs (Brown University field trip).	03/24
Azores, Portugal (Pico, Faial, São Miguel and Terceira): sampling of volcanic and hydrothermal environments, funded by SNSF. Ferrihydrite samples from Terceira were published in <i>Nature Communications</i> in 2025.	11/23
Iceland: adenosine triphosphate (ATP) analysis and collection of volcanic tephra, Nordic Astrobiology Network summer school.	06/16